Hey look we have a cluster now isn't that cool what does it do does it do stuff yes it does it's cool

Doing cluster stuff for fun and ~~profit~~ science

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BioWiMium

We have a new cluster

You might have *heard*

Time to give it an introduction! We'll cover:

- What is it
- How does it do stuff
- How do **you** do stuff on it
- I know it's annoyongly noisy oh glob I know stop throwing stuff 😭

What is a cluster
And why should you care

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- TermServ, Bertha
- Run looong tasks unsupervised
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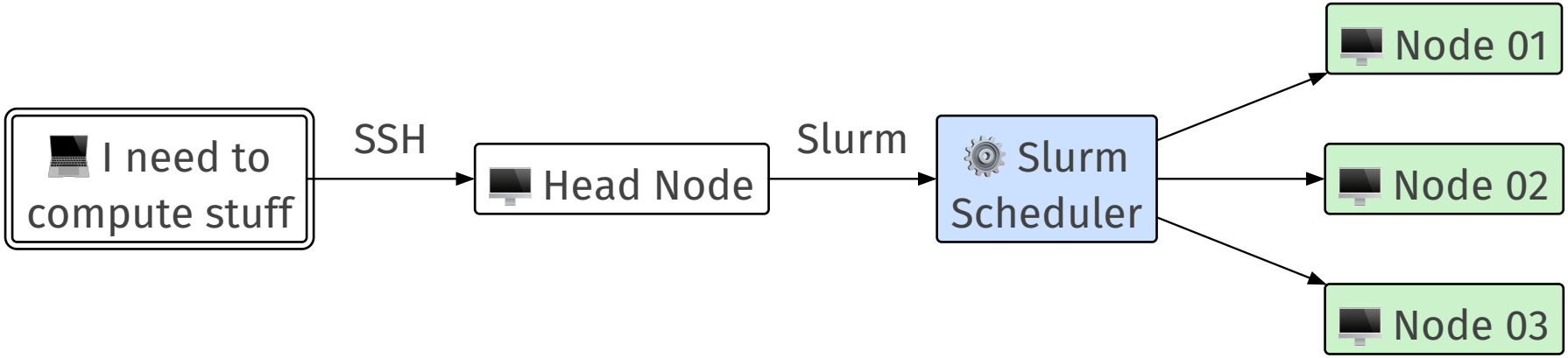
HPC (“Cluster”)

- Many powerful computers
- Access indirectly (head node)
- Shared (job management)
- Convenience: You’ll get there

What does it look like?
Kind of like this

Cluster topology

It only looks weird because it is



Terminology

Just to make sure you're confused

- The cluster has 12 compute **nodes** (self-contained computer)
  - Each node has 1 **CPU** (physical processor)
  - Each processor is made up of independent 96 **cores**
  - Each core can handle 2 **threads** (*not* independent)

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⚠️ **Slurm**: When you want 10 **threads** you request 10 “cpus”

Logging in

Slurm manages resources on the cluster

Connect to the cluster **head node** via SSH in one of two ways

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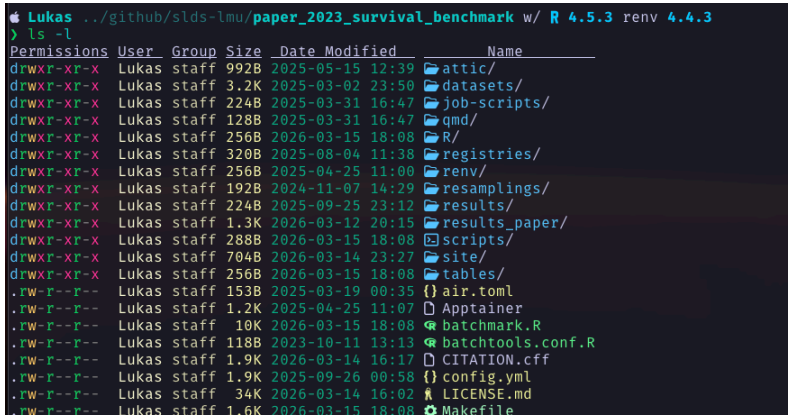
- Definitely cool and normal
- 1337 h4xX0r feeling



```
Lukas ../github/slds-lmu/paper_2023_survival_benchmark w/ R 4.5.3 renv 4.4.3
❯ ls -l
Permissions User  Group Size  Date Modified    Name
drwxr-xr-x  Lukas staff 992B 2025-05-15 12:39 attic/
drwxr-xr-x  Lukas staff 3.2K 2025-03-02 23:50 datasets/
drwxr-xr-x  Lukas staff 224B 2025-03-31 16:47 job-scripts/
drwxr-xr-x  Lukas staff 128B 2025-03-31 16:47 qmd/
drwxr-xr-x  Lukas staff 256B 2026-03-15 18:08 R/
drwxr-xr-x  Lukas staff 320B 2025-08-04 11:38 registries/
drwxr-xr-x  Lukas staff 256B 2025-04-25 11:00 renv/
drwxr-xr-x  Lukas staff 192B 2024-11-07 14:29 resamplings/
drwxr-xr-x  Lukas staff 224B 2025-09-25 23:12 results/
drwxr-xr-x  Lukas staff 1.3K 2026-03-12 20:15 results_paper/
drwxr-xr-x  Lukas staff 288B 2026-03-15 18:08 scripts/
drwxr-xr-x  Lukas staff 704B 2026-03-14 23:27 site/
drwxr-xr-x  Lukas staff 256B 2026-03-15 18:08 tables/
.rw-r--r--  Lukas staff 153B 2025-03-19 00:35 {} air.toml
.rw-r--r--  Lukas staff 1.2K 2025-04-25 11:07 Apptainer
.rw-r--r--  Lukas staff  10K 2026-03-15 18:08 batchmark.R
.rw-r--r--  Lukas staff 118B 2023-10-11 13:13 batchtools.conf.R
.rw-r--r--  Lukas staff 1.9K 2026-03-14 16:17 CITATION.cff
.rw-r--r--  Lukas staff 1.9K 2025-09-26 00:58 {} config.yml
.rw-r--r--  Lukas staff  34K 2026-03-14 16:02 LICENSE.md
.rw-r--r--  Lukas staff 1.6K 2026-03-15 18:08 Makefile
```

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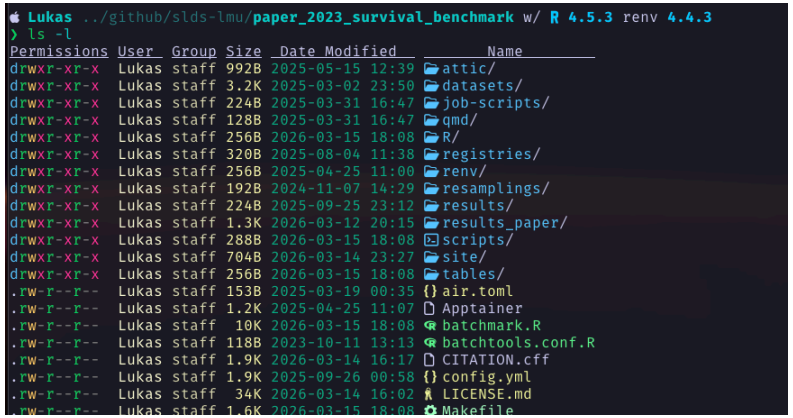

VScode / Positron

- Uses SSH for communication with head node
- GUI like on your own device
- All content is on the head node
- → Edit files conveniently
- Terminal / R console **runs on head node**

How are resources allocated?

Slurm manages resources on the cluster

- Example: You need **20 threads** for **6 hours** and **4GB of RAM** for `analysis.R`

Interactive

```
salloc --cpus-per-task=20 --mem=4G --time=06:00:00
```

```
salloc: Granted job allocation 137031
salloc: Nodes node01 are ready for job

# start R session
R

# Run your script
source("analysis.R")
```

Batch job

- Write an `analysis.sh` wrapper script

```
#!/bin/bash
#SBATCH --job-name=my-analysis
#SBATCH --output=job_%j.out
#SBATCH --cpus-per-task=20
#SBATCH --mem=4096M
#SBATCH --time=06:00:00

Rscript analysis.R
```

- Run `sbatch analysis.sh` and go home

Thank you for your attention!

www.leibniz-bips.de

Contact

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